Aayush Kumar JVS

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Summary

- AI/ML and Full-Stack Engineer with 5+ years of experience in GIS platforms, cloud systems, AI research, and scalable web applications, including engineering support for Apple Maps tooling at RMSI. Experienced in building production-grade solutions using Python, Vue.js, AWS, GraphQL, and CI/CD pipelines. Published researcher and PyCon US speaker specializing in efficient large language model deployment, quantization, and local inference workflows.

Experience

- **Google Summer of Code**
  Software Engineer — *May 2023 – Sept 2023*
  - **Google-Sponsored Project** Contributed to Red Hen Annotator Tool v2.0 in collaboration with Red Hen Labs through Google Summer of Code.
  - **Annotation Systems** Enhanced audio annotation workflows and expanded usability of existing tooling.
  - **AI Tooling** Built components leveraging NVIDIA NeMo Guardrails for annotation and workflow automation.
- **RMSI Pvt Ltd**
  Software Engineer (GIS) — *Aug 2020 – Aug 2021*
  - **GIS Platform Development** Built GIS tooling integrated with Confluence and Jira, improving spatial-data operational workflows by 35%.
  - **CI/CD Engineering** Designed CI/CD pipelines supporting Apple Maps tooling, accelerating deployment workflows by 40%.
  - **Leadership** Mentored 20+ engineers on production workflows and issue resolution processes.
  - **API & SDK Integration** Enhanced GIS systems using REST APIs, ArcGIS, FusionX, and internal SDK integrations.

Projects

- Encrypted Peer-to-Peer File Sharing System
  Research Project — *Jan 2023 – Mar 2023*
  - **Security Engineering** Built a decentralized encrypted peer-to-peer file-sharing system using Python sockets and cryptographic primitives.
- Analysis of Secured Federated Learning
  Research Project — *Apr 2023 – Jun 2023*
  - **Federated Learning** Analyzed tradeoffs between privacy, performance, and latency in secure aggregation protocols across distributed systems.
- Master's Thesis – Quantized LLMs on Mental Health Datasets
  Research Project — *Feb 2024 – Nov 2024*
  - **Model Benchmarking** Benchmarked quantized LLMs including QLoRA, BitsAndBytes, GGUF, and GGML using IMHI datasets to evaluate compression-versus-accuracy tradeoffs.

Skills

- **Programming Languages** Python, Java, JavaScript, TypeScript, Go, HTML5, CSS3
- **Frameworks & Libraries** PyTorch, TensorFlow, Keras, Vue.js, React Native, Angular, Node.js, Next.js, Django
- **Cloud & DevOps** AWS (EC2, Lambda, S3, ECS, DynamoDB, RDS), Microsoft Azure, Docker, Kubernetes, CI/CD, Git, Apache Kafka
- **Databases** PostgreSQL, MySQL, MongoDB, Redis, Oracle, GraphQL
- **Data Engineering & Analytics** Pandas, NumPy, PySpark, Hadoop, ArcGIS, FusionX
- **Visualization** Power BI, Tableau, D3.js, Plotly, GIS Mapping
- **Methodologies** Agile, Scrum, DevOps, Test-Driven Development, Root Cause Analysis

Publications

1. **Aayush Jannumahanti**. *Quantized Large Language Models for Mental Health Applications*: Benchmark study on efficiency, accuracy, and resource utilization.
2. **Aayush Kumar**. *Handwritten Recognition System Using Neural Networks and Guided Inputs.*
3. **Aayush Kumar**. *Symbolic Aggregate approXimate (SAX): An Innovative Outlook to Time Series Representation* — Gold Medal, Best Student Research; accepted at IJSER.
4. Venkatesh R., P. Yadhu, **Aayush Kumar**, Projjal Gupta. *Developing an Integrative Framework of Artificial Intelligence and Blockchain for Augmenting Smart Governance* — Silver Medal, Best Student Research.

Talks & Presentations

- **PyCon US 2026** — Presented a technical talk on *Running Large Language Models Efficiently on Laptops Using Python and Consumer Hardware.* Covered practical workflows for local inference, quantization, memory optimization, and deployment of open-source LLMs on resource-constrained systems using Python tooling.

Education

- Master of Science – Computer Science
  University of Maryland, Baltimore County (UMBC) — GPA: 3.4 *Aug 2022 – Dec 2024*
- Bachelor of Technology – Information Technology
  SRM Institute of Science and Technology — GPA: 3.0 *Jul 2016 – Jun 2020*